Who Gets the Credit?

Cross-Gender Model Transfer and Player Valuation in Women's Soccer

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Abstract

Introduction:

Player valuation models are often used in scouting and talent identification. A useful model should not only predict events well. It should also rank players in a stable and meaningful way. This paper studies whether models trained on men's soccer data assign credit to women's players in the same way as models trained on women's data.

Previous cross-gender studies have mostly focused on shot probability or local differences in event value. We extend this question to player rankings. We compare five valuation frameworks: xG, xA, xT, VAEP, and Atomic-VAEP. We pay particular attention to non-shooting actions, including interceptions, ball receptions, and progressive passes. These actions are important for defenders and hybrid-role players, but they may be valued differently across training populations.

Methods:

We use StatsBomb Open Data: 771 women's matches from 2023/24 and 1,517 men's matches from 2015/16. The time gap is a limitation, but it reflects the limited public availability of women's match event data. Within each metric, we train separate men's and women's models. Both models then score the same held-out women's events using match-level train-test splits. For xG, xA, VAEP, and Atomic-VAEP, we downsample the men's training data to match the number of women's events. Sample sizes vary by metric because different event types occur at very different rates, so cross-metric comparisons should be read with caution.

Results:

Most metrics show high ranking stability. The correlations between men's-trained and women's-trained player rankings are 0.977 for xG, 0.971 for xA, 0.975 for xT, and 0.954 for VAEP. Atomic-VAEP is the exception, with a much lower correlation of 0.652. The instability is strongest for defensive positions. Atomic-VAEP correlations are 0.365 for centre-backs and 0.413 for fullbacks and wingbacks, compared with 0.877 for forwards. All correlations are statistically significant at $p < 0.001$, but the key result is the size of the ranking shift under Atomic-VAEP.

Action-level decomposition shows that interceptions and receptions drive much of this defender instability. Progressive passes matter less than expected. This suggests that cross-gender sensitivity is concentrated in how models value defensive transitions, rather than in attacking progression alone. Women-specific training also does not always improve predictive performance. For example, xG Brier scores are nearly identical for men's-trained and women's-trained models.

Conclusion:

These findings matter for recruitment. Models appear more transferable for shot-driven roles, but they require caution for defenders and players whose value comes from transitional actions. A player such as Yamamoto Yuzuki, a Japanese international who moved from forward to right-back in 2025, shows the type of profile that may be affected. Her new role places more weight on interceptions and progression, exactly the kinds of actions where model transfer is least stable. Because the data come from one provider and from different time periods, these results should be read as a first step rather than a final estimate of gender effects. Future work should test the same question with synchronized men's and women's data from the same seasons.

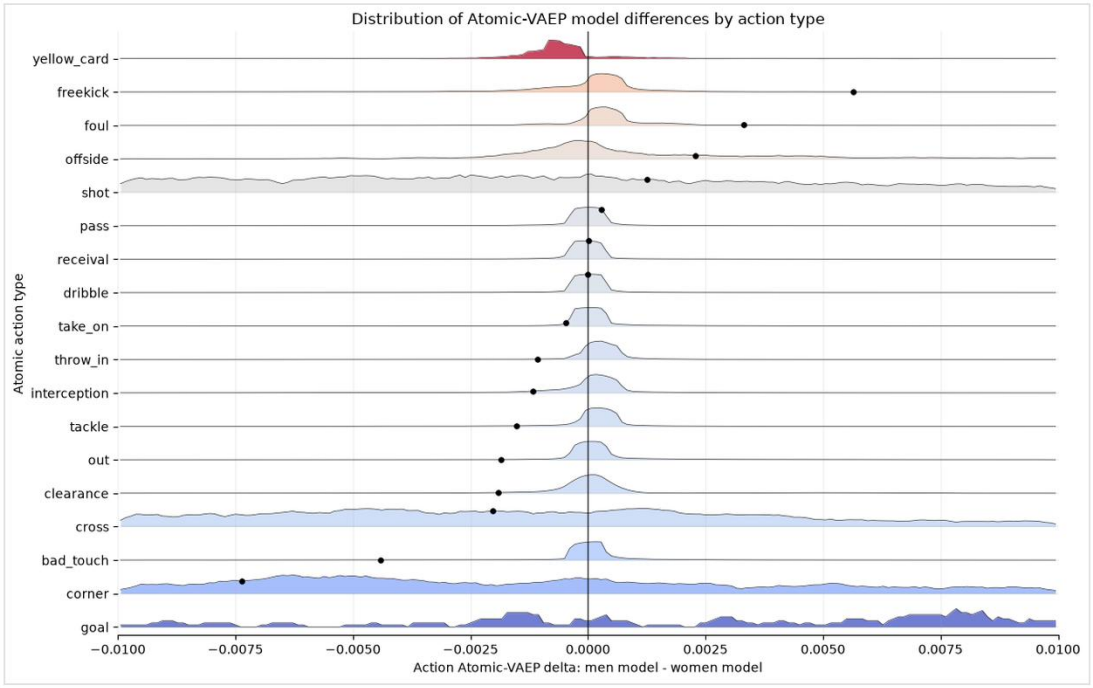


Figure 1: Action-type differences in Atomic-VAEP values between men-trained and women-trained models

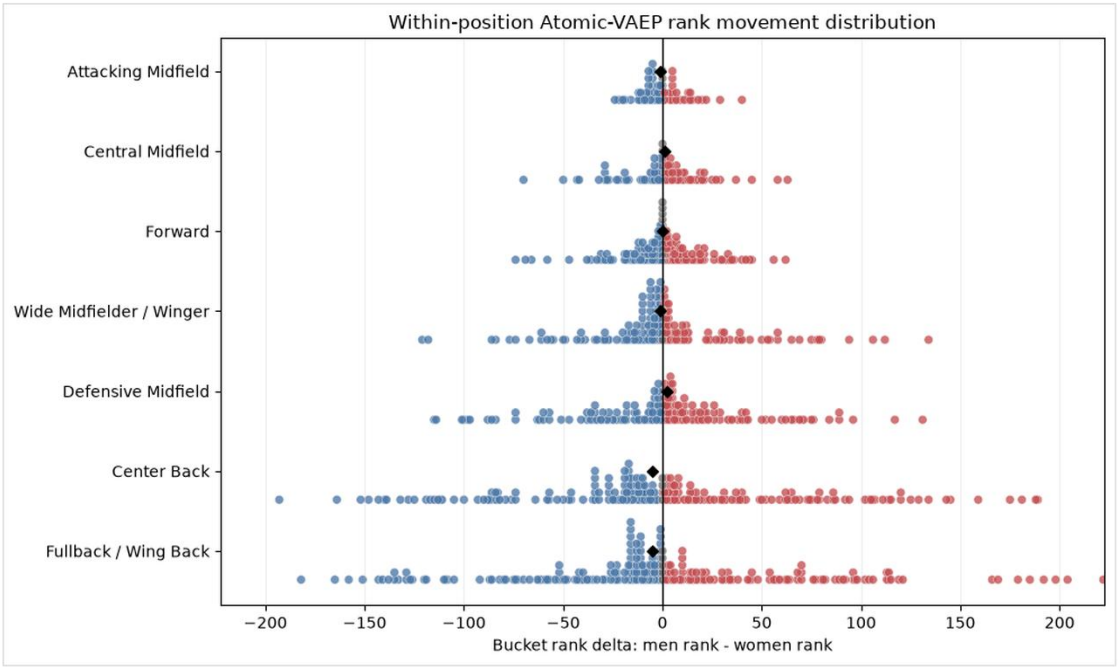


Figure 2: Position-specific ranking shifts for Atomic-VAEP between men-trained and women-trained models